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Summary

Kingdom Animalia comprises of a broad range of animal species, from tiny parasitic nematodes to the largest mammal the blue whale. The basic fundamental features such as levels of organisation, diploblastic and triploblastic organisation, patterns of symmetry, coelom, segmentation and notochord have enabled us to broadly classify the animal kingdom. Besides the fundamental features, there are many other distinctive characters which are specific for each phyla or class.

Animals are broadly classified into invertebrates and chordates. The animals which lack vertebral column are called invertebrates. The chordates are characterized by the presence of notochord, solid ventral nerve cord and gill slits. Kingdom Animalia are classified into eleven animal phyla as Porifera, Cnidaria, Ctenophora, Platyhelminthes, Aschelminthes, Annelida, Arthropoda, Mollusca, Echinodermata, Hemichordata and Chordata. Chordata is the largest phylum with three sup phyla Urochordata, Cephalochordata and Vertebrata. Subphylum Vertebrata includes two divisions, Agnatha and Gnathostomata. Agnatha comprises of the class Cyclostomata. Gnathostomata includes jawed fishes (Pisces) and Tetrapoda which includes the classes amphibia, reptilia, aves and mammals.

Glossary

Alternation of generation – Alternation of haploid sexual and diploid asexual generation in the life cycle of an animal.

Autonomy – Breaking of a body part.

Dioecious – Animals in which male and female reproductive organs occur in separate individuals.

Hermaphrodite – Animals with both male and female reproductive organs.

Mersentery – A thin double walled epithelial membrane that support alimentary canal and other organs in the abdominal cavity.

Regeneration – Act of growing a new body part which has been injured or lost.

Evaluation

1. The symmetry exhibited in cnidarians is
   a. Radial b. Bilateral
   c. Pentamerous radial d. Asymmetrical
2. Sea anemone belongs to phylum
   a. Protozoa b. Porifera
   c. Coelenterata d. Echinodermata
3. The excretory cells that are found in platyhelminthes are
   a. Protonephridia b. Flame cells
   c. Solenocytes d. All of these
4. In which of the following organisms, self fertilization is seen.
   a. Fish b. Round worm
   c. Earthworm d. Liver fluke
5. Nephridia of Earthworms are performing the same functions as
   a. Gills of prawn
   b. Flame cells of Planaria
   c. Trachea of insects
   d. Nematoblasts of Hydra